



AI DRIVEN CYBER SECURITY THREAT PREDICTION PLATFORM



VIRTUAL INTERNSHIP 6.0



ABOUT PROJECT

The AI-Driven Cybersecurity Threats Prediction Platform aims to enhance digital security using machine learning and real-time analytics.

By monitoring network traffic, the system identifies anomalies and predicts potential cyber threats before they occur.

The platform integrates an AI chatbot for user interaction and explainability.



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GOALS

Detect and predict potential cybersecurity threats in real-time.

Automate threat detection using AI/ML models.

Provide explainable AI insights for security analysts.

Enable human-AI interaction through an integrated chatbot.

Build scalable deployment using Kafka, APIs, and Streamlit dashboards.



ROADMAP OF PROJECT

Data Collection & Analysis

Import, clean, and analyze network traffic data.
Identify anomaly indicators (e.g., unusual IPs, high packet rate).



1



Research & Planning

Define project goals, dataset, and architecture.
Choose the dataset (CIC-IDS2017 or UNSW-NB15).

2

3



Model Development (Phase 1)

Select and train baseline anomaly detection models.
Split dataset (Train: 80%, Test: 20%).

4



Model Optimization

Tune hyperparameters and add explainability features.
Save best-performing model (model.pkl).

5



Real-Time Threat Detection

Simulate live data stream and create alert system.
using socket or Kafka.

6



Chatbot Integration

Build AI chatbot for security explanations and commands.

7



Integration & Deployment

Combine all modules and deploy prototype.

8

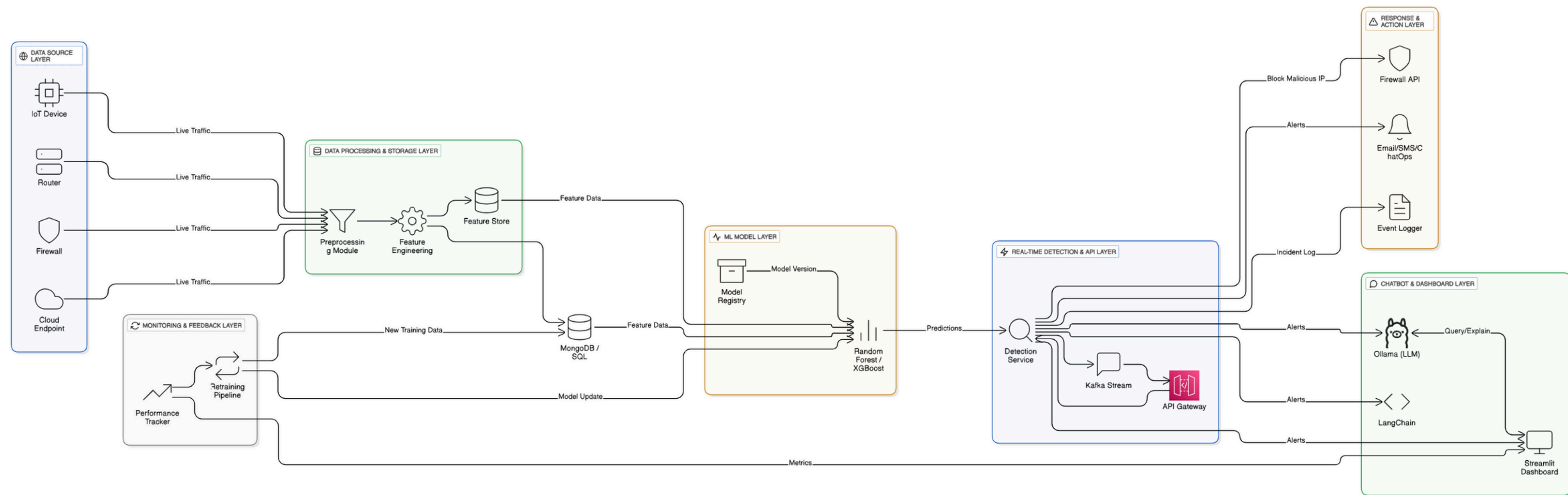


Testing & Final Report

Evaluate system, create demo, and prepare submission



ARCHITECTURE





KEY CHALLENGES



Real-Time Data Processing from Multiple Sources :

Handling live traffic from IoT devices, routers, firewalls, and cloud endpoints generates high-velocity data.



Maintaining Model Accuracy Over Time

Changing attack patterns and network behaviors degrade ML model performance.



Explainability & Human-AI Collaboration

Security analysts need clear explanations behind AI-driven alerts.

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FUTURE SCOPE & ENHANCEMENT

SCOPE

Real-time monitoring of IoT, routers, and firewalls.

Automated threat blocking and reporting.

Integrated chatbot with explainable AI for queries.

ENHANCEMENTS

Integration with cloud-based security intelligence.

Deep learning models for advanced anomaly detection.

Multi-cloud deployment and continuous model retraining.



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THANK YOU!

